Policy-Grounded Explainable AI for Medicare Fraud Detection Integrating Machine Learning and Retrieval-Augmented Generation

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ABSTRACT

Medicare fraud results in billions of dollars in annual losses for the United States Medicare system. Although machine learning models can identify suspicious claims and providers at scale, their limited transparency often undermines investigator trust and real-world adoption. This research proposes a guidance-grounded, explainable artificial intelligence framework for Medicare fraud detection that integrates predictive machine learning with policy-aware language-based interpretation. The framework aligns model outputs with valid Medicare policy and regulatory guidance from the Centers for Medicare & Medicaid Services (CMS) and the Office of Inspector General (OIG), supporting investigators' work more openly and with clear evidence.

The proposed approach develops explainable fraud detection models that address class imbalance and generate risk scores accompanied by feature-level explanations using Shapley Additive Explanations (SHAP). It further examines the use of a large language model, GPT-4, augmented with retrieval-augmented generation (RAG), to provide guidance-grounded interpretation of machine learning outputs. Explanations are delivered through a chatbot-based interface that enables interactive, human-in-the-loop fraud investigation. The framework is evaluated against traditional model-only methods in terms of detection performance, interpretability, and investigator usability.

Keywords: Medicare Fraud Detection, Large Language Models, Generative AI, SHAP, RAG

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1. BACKGROUND

Medicare is one of the largest healthcare programs in the United States, and it is so large and complex that it is at high risk of fraud, waste, and abuse. Every year, fraudulent billing results in billions of dollars in incorrect payments, putting pressure on the Medicare system and eroding people's trust. It is hard to detect Medicare fraud because fake claims are rare, often look like legitimate bills, and are hidden among large volumes of normal claims. To address this problem, agencies such as the Centers for Medicare & Medicaid Services (CMS) and the Office of Inspector General (OIG) use audits, expert review, and automated systems to detect problems. Computer programs that learn from data are especially helpful because they can analyze large volumes of claims and identify unusual provider actions faster than people can. However, many machine learning models operate as black boxes, producing risk scores without clearly explaining their reasoning.

This lack of clarity is a big problem in Medicare fraud investigations. Investigators must explain their decisions with clear evidence that meets official Medicare requirements. If the results from computer models are hard to understand or do not match the language of the rules, people are less likely to trust or use them, even if the models are very accurate.

Explainable artificial intelligence methods help address this issue by highlighting the features that influence model predictions. While these techniques improve transparency, they often remain technical and statistical. In highly regulated domains such as healthcare, explanations must be grounded in policy guidance to support real investigative decision-making.

Recent advances in large language models offer new ways to understand and use regulatory documents at scale. When these models are used with tools that can find information, they can link analysis results to official CMS and OIG policy sources. However, current approaches to detecting Medicare fraud rarely combine predictive modeling, clear explanations, and policy understanding in a single system.

This study examines how to combine different approaches to detecting Medicare fraud to yield results that are accurate, easy to understand, and useful. By linking machine learning predictions with explainable methods and authoritative Medicare policy and regulatory guidance, the framework aims to help investigators understand why a provider is flagged and make more informed decisions about which cases to focus on.

2. INTRODUCTION

Medicare fraud remains one of the most financially damaging and persistent challenges in the United States (CMS, 2023). Among its many forms, Medicare fraud is particularly complex and large-scale, resulting in billions of dollars in improper payments each year (OIG, 2023). The Centers for Medicare & Medicaid Services (CMS) and the Office of Inspector General (OIG) continue to invest heavily in systems to detect and recover fraudulent expenditures. However, given the enormous volume of claims and the constantly evolving nature of fraud schemes, traditional audit and rule-based systems are inefficient, reactive, and limited in scalability (Bauder & Khoshgoftaar, 2018). Machine learning (ML) has improved fraud detection by analyzing large datasets and uncovering hidden patterns. However, many ML models are hard to interpret. They often provide risk scores without explaining why a claim was flagged or how the prediction aligns with Medicare policy and regulatory guidance (Carvalho et al., 2019). This lack of clarity creates two main problems for investigators: trust and actionability (Adadi & Berrada, 2018; Guidotti et al., 2018; Doshi-Velez & Kim, 2017), as investigative decisions must be supported by clear, auditable regulatory evidence. As a result, even high-performing machine learning models are not widely adopted in real-world fraud investigations. The primary research gap lies in the absence of a unified framework that connects policy-aware reasoning with data-driven fraud detection (Hancock et al., 2024; Bounab et al., 2024; Lewis et al., 2020). Most prior studies address individual challenges, such as handling class imbalance or improving model transparency using techniques like SHAP, but rarely integrate these components into a single, end-to-end system.

This proposal introduces a policy-aware, explainability-methods framework for Medicare fraud detection, designed to improve transparency and investigator usability. The approach combines predictive machine learning models with explainability methods, particularly Shapley Additive Explanations (SHAP), to identify high-risk claims and

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providers and to clarify how individual features contribute to fraud risk predictions. This proposal presents a framework for Medicare fraud detection that aims to make the process more transparent and easier for investigators to use. The system has two components. The first component uses machine learning models and explainability tools, such as Shapley Additive Explanations (SHAP), to identify high-risk claims and providers and show which features influence risk predictions. The second component, it uses large language models (LLMs) with retrieval-augmented generation (RAG) to pull in relevant guidance from official CMS and OIG sources, helping investigators interpret fraud risks based on policy. The machine learning models highlight risky claims and providers, while the LLMs connect these results and SHAP explanations to the right regulatory guidance. Investigators can use a chatbot interface to explore fraud signals, identify contributing factors, and review supporting policy references. By combining these technologies, the system turns complex model outputs into clear, policy-based insights that help investigators make better decisions.
This research is designed to:

a) Build a Medicare fraud detection model that not only predicts fraud risk for claims and providers but also uses SHAP to clearly show which factors lead to those predictions.
b) Develop and evaluate a framework that uses large language models (LLMs) with RAG to incorporate trusted Medicare policy and regulatory guidance and combine these explanations with machine learning outputs within a chatbot-based interface to support Medicare fraud investigation workflows.

3. RELATED WORK

Detecting fraud in Medicare and healthcare systems has been widely studied. Prior research aligns with two themes relevant to this study: (1) data-level balancing and ML-based detection and (2) AI- and LLM-based explainability.

3.1 Use of Data-Level and Machine Learning in Medicare Fraud Detection

Hancock et al. (2024) present data reduction techniques for highly imbalanced Medicare Big Data. In their study, they used two datasets synthesized from two US government agencies, the CMS and the United States OIG. Their datasets are derived from publicly accessible Medicare data for the years 2013 to 2019. Hancock et al. assess the combined efficacy of two data reduction techniques: Random Under-sampling (RUS) and a novel ensemble supervised feature selection method. The techniques are applied to optimize Machine Learning models for fraud identification in the classification of highly imbalanced Big Medicare Data. They built and compared several models, including Logistic Regression, Random Forest, CatBoost, LightGBM, XGBoost, and Extremely Randomized Trees, under five experimental scenarios: RUS, Feature selection, Feature selection followed by RUS, RUS followed by feature selection, and No reduction. The best performance was achieved when both RUS and feature selection were combined, outperforming models that used all available data or features.
Bounab et al. (2024) address the challenge of imbalanced datasets, which make healthcare fraud detection, particularly within the Medicare Part B dataset, difficult. To mitigate this issue, the authors generate synthetic instances by replicating existing minority-class samples and extracting categorical features to increase diversity among fraudulent cases. At the data level, they apply a hybrid resampling technique, SMOTE-ENN, which combines Synthetic Minority Over-sampling Technique (SMOTE) with Edited Nearest Neighbors (ENN) to balance the dataset better. Six ML models are then trained to classify claims, and their performance is evaluated using accuracy, F1-score, recall, precision, and the AUC-ROC metric. The results demonstrate that the Decision Tree model outperformed all other classifiers, achieving a remarkable 0.99 score across all evaluation metrics.
Bauder and Khoshgoftaar (2018) used Medicare Part B data from 2012–2015 and matched it with the OIG List of Excluded Individuals/Entities (LEIE) to label known fraudulent providers. To address the severe class imbalance, the authors created seven class distributions and evaluated the behavior and fraud detection performance of six machine learning algorithms. The results showed that the Random Forest (RF100) model with a 90:10 class distribution achieved the best performance, reaching an AUC of 0.873, demonstrating its superior ability to detect fraudulent providers in highly imbalanced Medicare data.
While these studies improved predictive accuracy and class balance, they do not address explainability or policy-level interpretability. None provides insight into why models flag specific claims or how decisions align with

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Medicare policy. This research extends their work by integrating predictive modeling with retrieval-based reasoning through three innovations: a RAG-powered chatbot that grounds explanations in CMS and OIG documentation; explainable ML via SHAP and counterfactuals for transparency; and an integrated evaluation that combines interpretability with accuracy to produce policy-grounded decision support.

3.2 Use of AI and Large Language Models in Medicare Fraud Detection

Carcillo et al. (2018) addressed the challenge of credit card fraud detection in a streaming data environment, where transactions arrive continuously, and the cost of human investigation limits labeling. They proposed and evaluated multiple active learning (AL) strategies designed to improve fraud detection while minimizing labeling costs. They compared supervised, semi-supervised, and unsupervised AL methods under real-world conditions provided by Worldline, a European financial services company. The dataset included over 12 million transactions, of which only about 0.2% were fraudulent. The study emphasized the exploitation, exploration trade-off, meaning models must balance querying high-risk transactions to catch more fraud. The results include that combining exploration and exploitation yielded better long-term fraud detection than focusing solely on the highest-risk alerts. Also, SSSL strategies reduced labeling costs effectively while maintaining competitive accuracy.
Lewis et al. (2020) introduced a general-purpose fine-tuning framework for RAG models that combine parametric memory stored within a language model, and non-parametric memory (external, retrieved documents) to improve text generation. The authors compared two RAG formulations: one that conditions on the same retrieved passages throughout the generated sequence, and another that allows different passages to influence each generated token. Their model was fine-tuned on several knowledge-intensive tasks, including open-domain question answering, abstractive question answering, Jeopardy-style question generation, and fact verification. The results showed that RAG models generated text that was more specific, diverse, and factual than state-of-the-art parametric-only baselines. In particular, the RAG-Token variant achieved 42.7% higher factuality and 37.4% greater specificity in human evaluations compared to a BART-based baseline, demonstrating its effectiveness in producing grounded and knowledge-rich responses.
Although Carcillo et al. and Lewis et al. advanced automated and retrieval-based reasoning, their approaches have not been adapted to healthcare fraud contexts that require explainability and regulatory grounding. Carcillo's methods lack a natural-language interface, whereas Lewis's RAG model has not been applied to domain-specific policy references. Building on these insights, this study applies RAG and LLM frameworks to detect Medicare fraud. The Fraud Investigator Copilot introduces a human-in-the-loop chatbot that allows investigators to query, verify, and interpret model results while grounding explanations in CMS and OIG policies, thereby transforming fraud detection into an evidence-based, transparent, and policy-aligned workflow.

4. Methodology

4.1 Research Model – KDD Process Model

The Knowledge Discovery in Databases (KDD) process model serves as the research framework for this proposal (Figure 1). Rather than focusing on how theoretical factors interact, the KDD framework follows a structured, step-by-step, data-driven approach well-suited to large-scale Medicare fraud detection. The KDD process encompasses data selection, preprocessing, and transformation, model development, evaluation, and interpretation, thereby promoting methodological rigor, transparency, and reproducibility throughout the research workflow.

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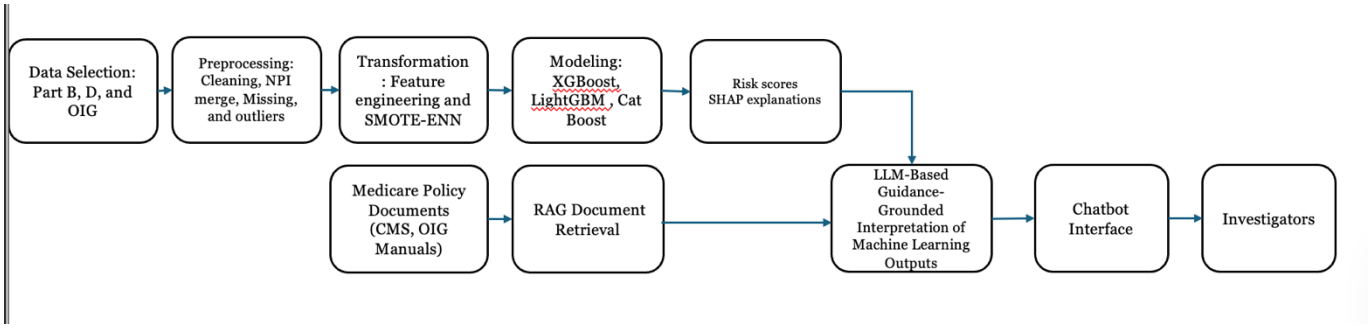


Figure 1: KDD process model. This figure illustrates the complete workflow, from data collection through SHAP interpretation to the LLMs for fraud investigation.

4.2 Data Type and Sources

This study combines structured Medicare claims data with official policy documents to build predictive models and interpret results in line with policy guidelines. The claims data come from CMS Part B and Part D files from 2013 to 2022, which include information about provider billing, services used, and prescriptions. This information is used to find patterns linked to fraud. Fraud labels are taken from the OIG List of Excluded Individuals and Entities (LEIE). The policy background comes from trusted Medicare sources, such as CMS program manuals and OIG fraud alerts, which are organized during data preparation to help explain the model results using policy rules. Using these data sources together enables the development of a clear, policy-based fraud detection system.

Table 1. Key variables used in the analysis and their roles in fraud labeling, modeling, and interpretation

Dataset	Variable	Description	Type	Role in KDD Process	Rationale for Fraud Detection
LEIE (OIG)	NPI	Provider ID	Key join field	Links LEIE with Part B and D	Ensures accurate provider-level integration across datasets.
Part B	HCPCS_C	Procedure code	IV	Core billing feature	Identifies services billed; certain codes associated with overutilization or abuse

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Part B	Rndrng_Prvr_Type	Specialty code	IV	Important structural feature for modeling	Specialties are associated with distinct service patterns and risk profiles.
Part B	Place_Of_Srvc	Facility vs non-facility billing	IV	Feature for modeling billing behavior	Useful in detecting false facility claims or abnormal site-of-service patterns.
Part B	Tot_Srvcs	Total services billed	IV	Feature creation (billing intensity)	High service volume relative to peers may indicate aggressive billing.
Part B	Tot_Benes	Number of unique beneficiaries	IV	Combined with Tot_Srvcs for anomaly ratios	Large discrepancies between services and beneficiary count may signify phantom billing.
Part B	Avg_Sbmtd_Chrg	Average submitted charge	IV	Feature for cost pattern analysis	Providers who consistently submit unusually high charges may be upcoding.
Part B	Avg_Mdcr_Pymt_Amt	Average Medicare payment	IV	Paired with Avg_Sbmtd_Chrg	The gap between submitted and paid amounts helps detect billing manipulation.
Part B	Avg_Mdcr_Alowd_Amt	Average Medicare allowed amount per service	IV	Payment level feature	Providers billing unusually high allowed amounts relative to peers may indicate upcoding, unnecessary procedures, or inflated billing practices.

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Part B	Avg_Mdcr_Stdzd_Amt	Average standardized Medicare payment amount	IV	Cost normalization featur	Removes geographic payment adjustments; high standardized payments suggest true service intensity rather than location effects, helping identify abnormal billing behavior.
Part B	Rndrng_Prvdr_RUCA	Provider rural–urban classification (RUCA code)	IV	Contextual adjustment feature	Accounts for practice location differences; unusual billing patterns in low-service rural areas or excessive services in specific geographic contexts may indicate fraud schemes or service concentration anomalies.
Part D	Brnd_Name	Brand (trade) name of the prescribed drug	IV	Drug selection pattern feature	Preference for costly brand drugs when generics exist may indicate kickbacks or non-medical prescribing behavior..
Part D	Gnrc_Name	Generic name of prescribed drug	IV	Drug category normalization	Enables detection of brand-over-generic prescribing patterns and therapeutic duplication schemes.
Part D	Tot_30day_Fills	Standardized 30-day prescription fills	IV	Prescription volume intensity	Extremely high refill frequency relative to peers may indicate automated refills or pill-mill behavior.
Part D	Tot_Clms	Total prescription claims submitted	IV	Core utilization feature	Abnormally high claim counts compared to specialty norms suggest excessive prescribing activity.
Part D	Tot_Day_Suply	Total days medication supplied	IV	Treatment duration feature	Excessively long medication supply periods may indicate chronic over-prescription or unnecessary therapy continuation.

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Part D	Tot_Drug_Cst	Total cost of prescribed drugs	IV	Financial risk indicator	Unusually high drug spending per provider or per claim may signal expensive drug schemes or reimbursement abuse.

4.3 Data Preprocessing and Preparation

After collecting the data, CMS Part B and Part D claims are combined with OIG exclusion records using the National Provider Identifier (NPI) to create provider-level fraud labels. The data are cleaned to fix missing information, inconsistent IDs, and unusual values. To address a large difference in the number of fraud and non-fraud cases, a specialized resampling method is used. Number-based data are adjusted to a common scale, and category-based data are converted to a format the model can use. At the same time, Medicare policy documents are split into sections and organized to help the system to provide policy and regulatory guidance.
explanations of model results.

4.4 Analytical Methods

Ensemble machine learning algorithms, such as LightGBM, XGBoost, and CatBoost, will be used for fraud prediction due to their proven effectiveness with large, sparse, and imbalanced Medicare datasets (Hancock et al., 2024). Model performance will be assessed using precision, accuracy, recall, F1-score, and ROC-AUC, which are established metrics for fraud detection in prior research (Iqbal et al., 2025). SHAP values and counterfactual explanations (Khan et al., 2025) will facilitate local interpretability and highlight the features that most significantly influence model predictions. In addition, policy-aware interpretation is supported by a large language model, specifically GPT-4, augmented with retrieval-augmented generation (RAG) to retrieve and incorporate relevant CMS and OIG policy and regulatory guidance documents to explain machine learning outputs. The same GPT-4 model is used for both guidance-grounded interpretation and interactive, conversational exploration of results within a chatbot interface. This integrated design supports transparent, evidence-based, and investigator-centered fraud analysis without influencing model predictions.

5. Expected Contributions

This research is expected to:

a) Show how clear and understandable machine learning methods, especially SHAP, can help find and explain fraud risk in Medicare claims and provider data.
b) Make fraud risk predictions clearer and help investigators trust them by connecting model explanations to official Medicare policy and regulatory guidance.
c) Offer a reusable analysis approach for data science projects that need to be clear, easy to check, and involve people in making decisions.

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